

How drugs act: molecular aspects

Targets for drug action

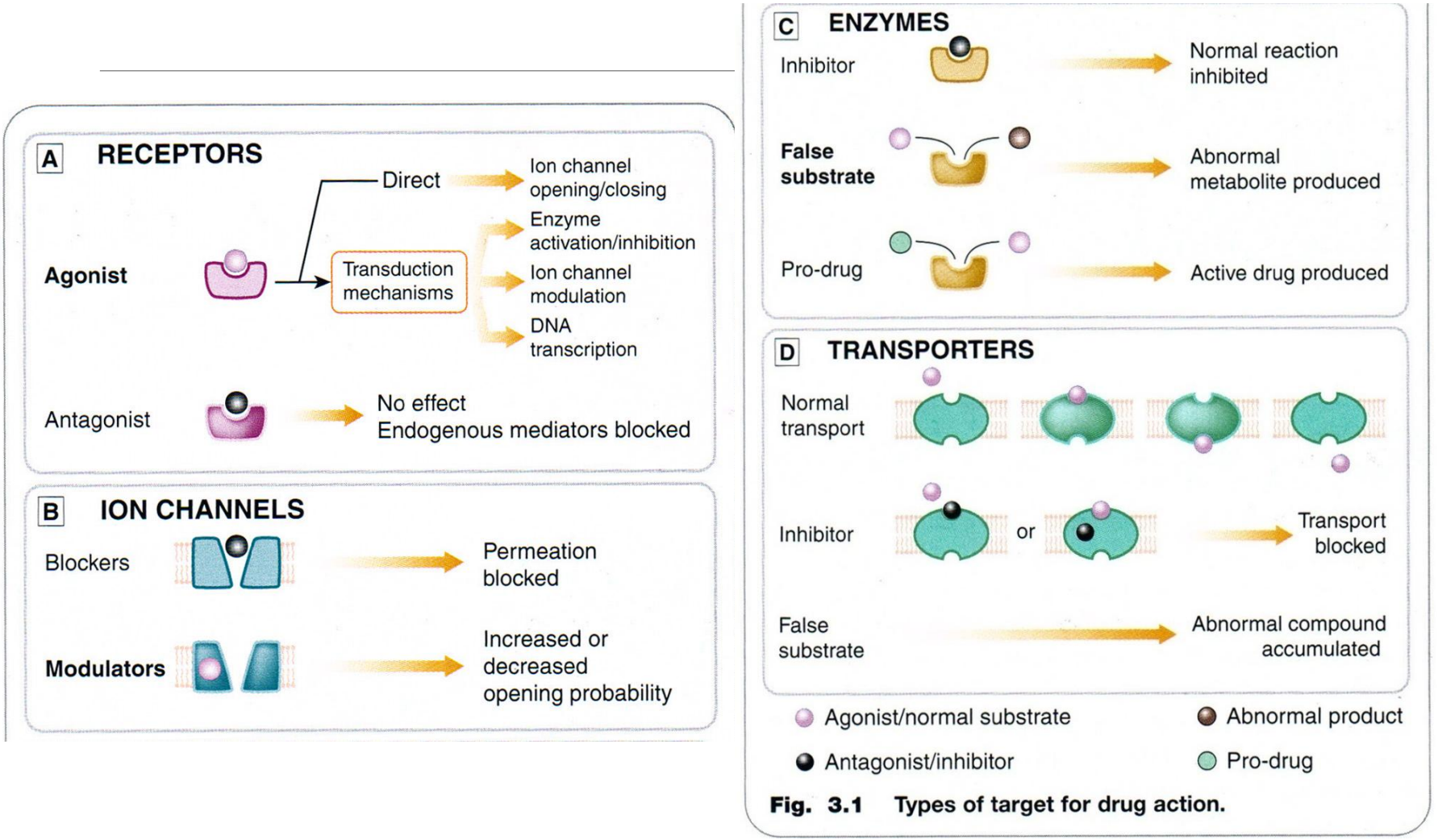


Fig. 3.1 Types of target for drug action.

Types of Receptor

1. **Ligand-gated ion channel**
2. **G-protein-coupled receptor**
3. **Kinase-linked receptor**
4. **Nuclear receptor**

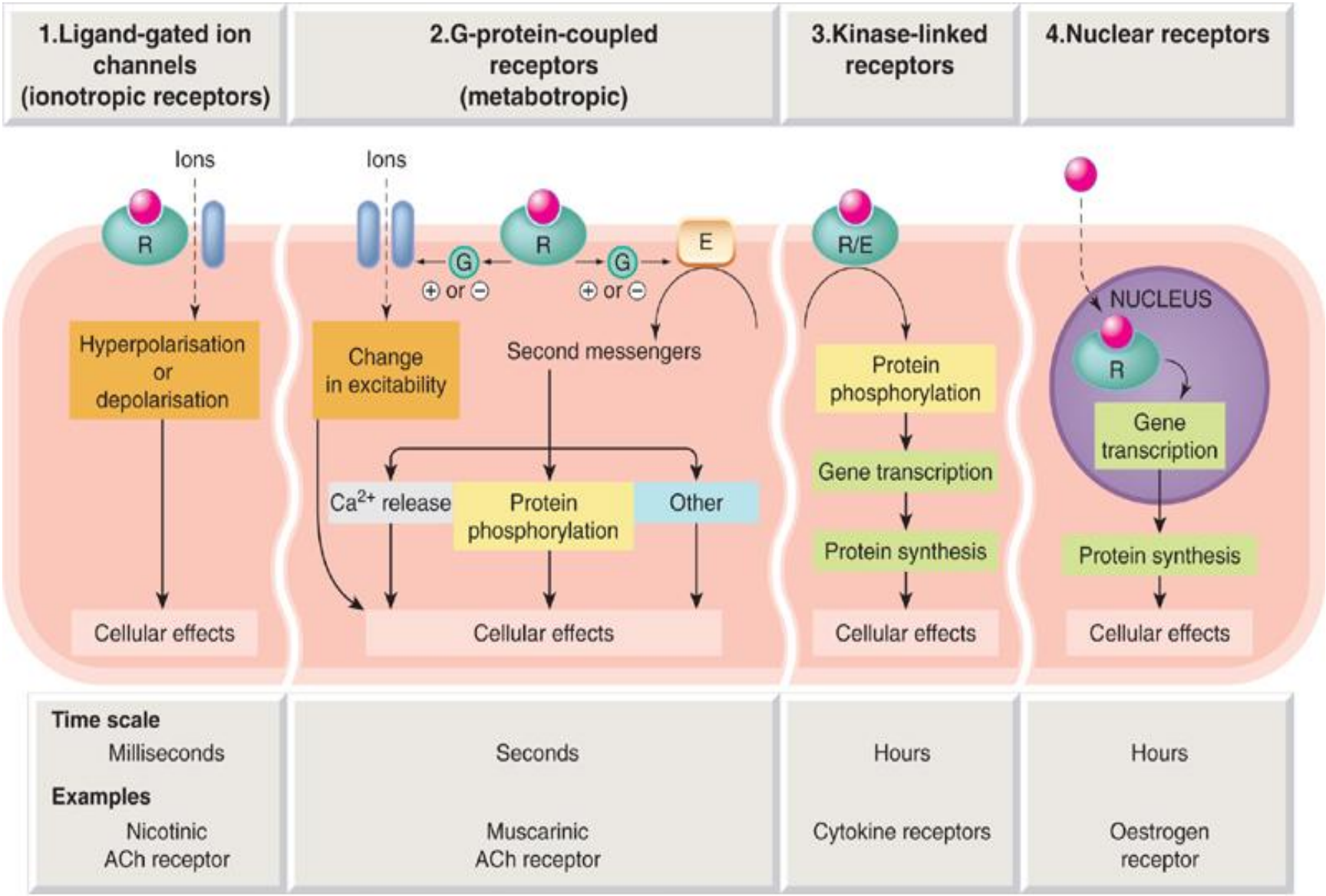


Fig. Types of receptor-effector linkage.
ACh, acetylcholine; E, enzyme; G, G-protein; R, receptor.

The four main types of receptor

	Type 1: ligand-gated ion channels	Type 2: G-protein-coupled receptors	Type 3: receptor kinases	Type 4: nuclear receptors
Location	**Membrane**	**Membrane**	**Membrane**	**Intracellular**
Effector	**Ion channel**	**Channel or enzyme**	**Protein kinases**	**Gene transcription**
Coupling	**Direct**	**G-protein or arrestin**	**Direct**	**via DNA**
Examples	**Nicotinic acetylcholine receptor, $GABA_A$ receptor**	**Muscarinic acetylcholine receptor, adrenoceptors**	**Insulin, growth factors, cytokine receptors**	**Steroid receptors**
Structure	**Oligomeric assembly of subunits surrounding central pore**	**Monomeric or oligomeric assembly of subunits comprising seven transmembrane helices with intracellular G-protein-coupling domain**	**Single transmembrane helix linking extracellular receptor domain to intracellular kinase domain**	**Monomeric structure with receptor- and DNA-binding domains**

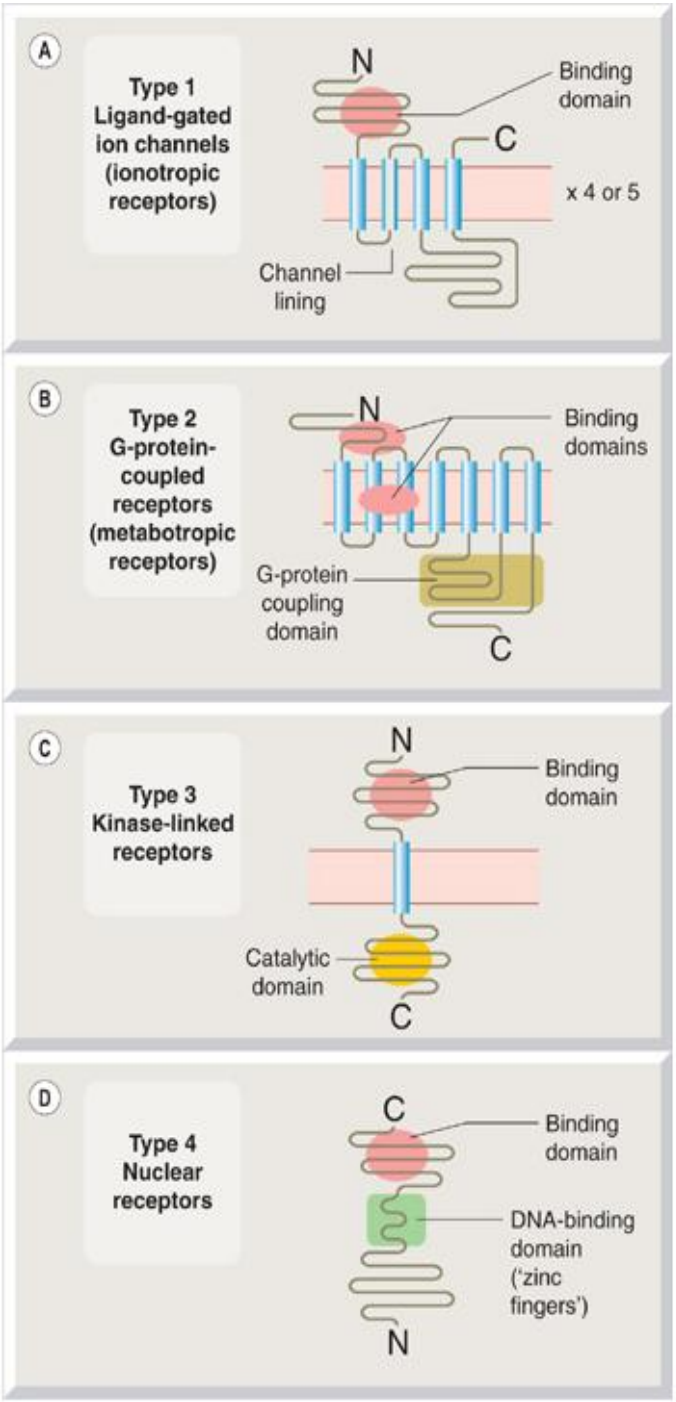


Fig. General structure of four receptor families.

The rectangular segments represent hydrophobic α-helical regions of the protein comprising approximately 20 amino acidse membrane-spanning domains of the receptors. **[A]** Type 1: ligand-gated ion channels. Many ligand-gated ion channels comprise four or five subunits of the type shown, the whole complex containing 16-20 membrane-spanning segments surrounding a central ion channel. **[B]** Type 2: G-protein-coupled receptors. **[C]** Type 3: kinase-linked receptors. Most growth factor receptors incorporate the ligand-binding and enzymatic (kinase) domains in the same molecule, as shown, whereas cytokine receptors lack an intracellular kinase domain but link to cytosolic kinase molecules. Other structural variants also exist. **[D]** Type 4: nuclear receptors that control gene transcription.

Ligand-gated ion channel

- **These are sometimes called ionotropic receptors.**
- **They are involved mainly in fast synaptic transmission.**
- **There are several structural families, the commonest being heteromeric assemblies of four or five subunits, with transmembrane helices arranged around a central aqueous channel.**
- **Ligand binding and channel opening occur on a millisecond timescale.**
- **Examples include the nicotinic acetylcholine, GABA type A ($GABA_A$), glutamate (NMDA) and 5- hydroxytryptamine type 3 ($5\text{-}HT_3$) receptors.**

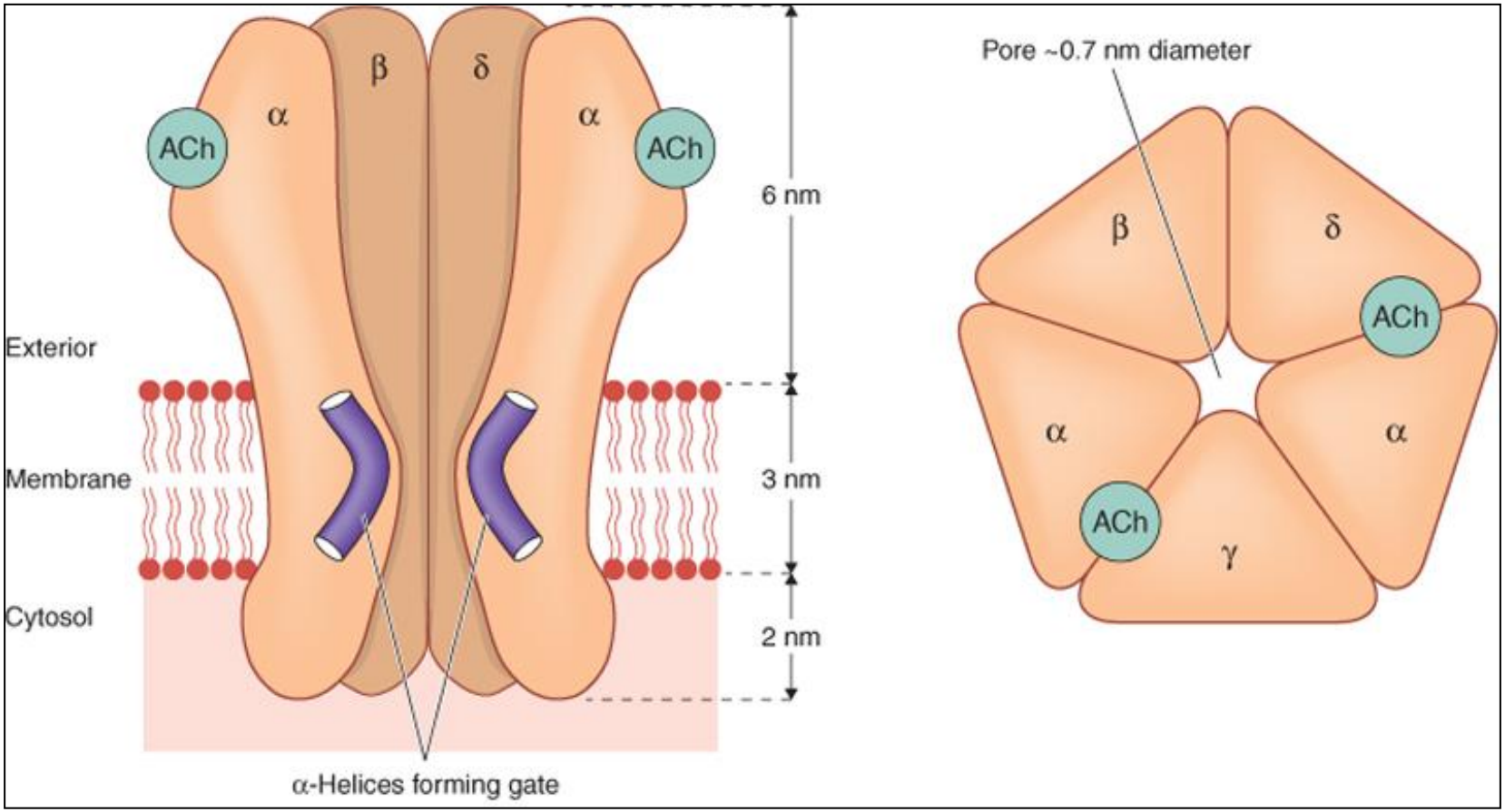


Fig. **Structure of the nicotinic acetylcholine receptor (a typical ligand-gated ion channel).**
The five receptor subunits (α_2, β, γ, δ) form a cluster surrounding a central transmembrane pore, the lining of which is formed by the M_2 helical segments of each subunit. These contain a preponderance of negatively charged amino acids, which makes the pore cation selective. There are two acetylcholine binding sites in the extracellular portion of the receptor, at the interface between the α and the adjoining subunits. When acetylcholine binds, the kinked α helices either straighten out or swing out of the way, thus opening the channel pore.

G-protein-coupled receptor

- **These are sometimes called metabotropic or seven-transmembrane-domain (7-TDM) receptors.**
- **Structures comprise seven membrane-spanning α-helices.**
- **The G-protein is a membrane protein comprising three subunits (α, β, γ), the α subunit possessing GTPase activity.**
- **The G protein interacts with a binding pocket on the intracellular surface of the receptor.**
- **When the G protein binds to an agonist-occupied receptor, the α subunit binds GTP, dissociates and is then free to activate an effector (e.g. a membrane enzyme). In some cases, the βγ subunit is the activator species.**

G-protein-coupled receptor

- **Activation of the effector is terminated when the bound GTP molecule is hydrolyzed, which allows the α subunit to recombine with βγ.**

- **There are several types of G-protein, which interact with different receptors and control different effectors.**

- **Examples include muscarinic acetylcholine receptors, adrenoceptors, neuropeptide and chemokine receptors, and proteinase-activated receptors.**

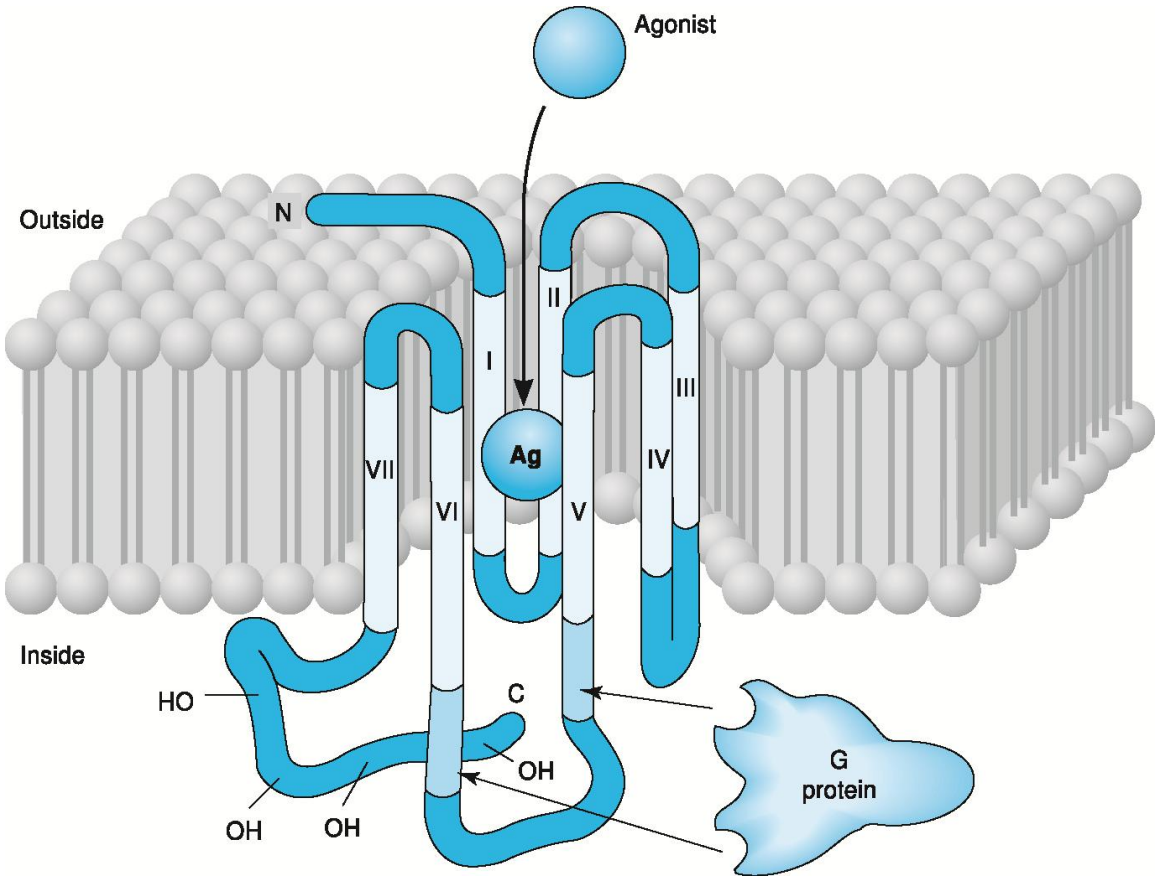


Fig. Transmembrane topology of a typical serpentine receptor.
The receptor's amino (N) terminal is extracellular (above the plane of the membrane), and its carboxyl (C) terminal intracellular. The terminals are connected by a polypeptide chain that traverses the plane of the membrane seven times. The hydrophobic transmembrane segments (light color) are designated by roman numerals (I-VII). The agonist (Ag) approaches the receptor from the extracellular fluid and binds to a site surrounded by the transmembrane regions of the receptor protein. G proteins interact with cytoplasmic regions of the receptor, expecially with protions of the third cytoplasmic loop between transmembrane regions V and VI. The receptor's cytoplasmic terminal tail contains numerous serine and threonine residues whose hydroxyl (-OH) groups can be phosphorylated. This phosphorylation may be associated with diminished receptor-G protein interaction.

G-protein-coupled receptor families

Family	Receptors	Structural features
A: rhodopsin family	The largest group. Receptors for most amine neurotransmitters, many neuropeptides, purines, prostanoids, cannabinoids, etc.	Short extracellular (N terminal) tail. Ligand binds to transmembrane helices (amines) or to extracellular loops (peptides)
B: secretin/glucagon receptor family	Receptors for peptide hormones, including secretin, glucagon, calcitonin	Intermediate extracellular tail incorporating ligand-binding domain
C: metabotropic glutamate receptor/calcium sensor family	Small group. Metabotropic glutamate receptors, $GABA_B$ receptors, Ca^{2+}-sensing receptors	Long extracellular tail incorporating ligand binding domain

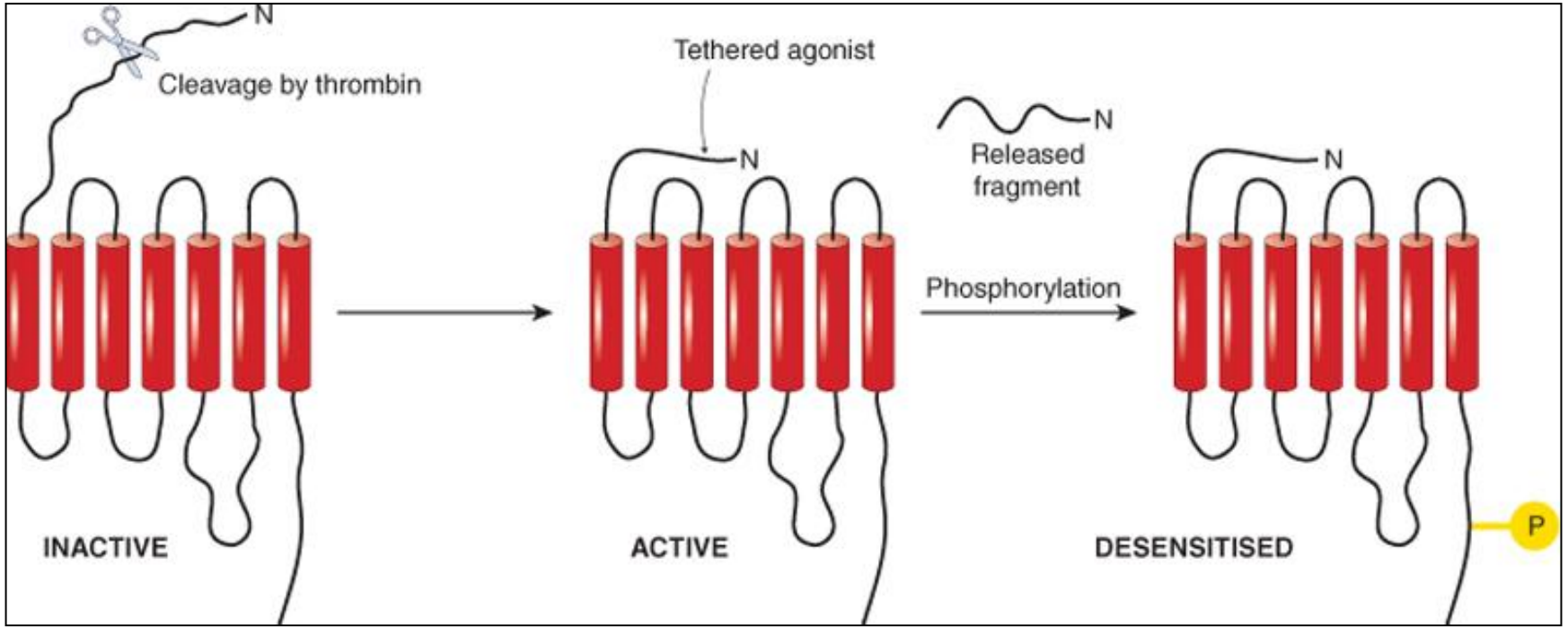


Fig. Activation of a proteinase-activated receptor by cleavage of the N-terminal extracellular domain.

Inactivation occurs by phosphorylation. Recovery requires resynthesis of the receptor.

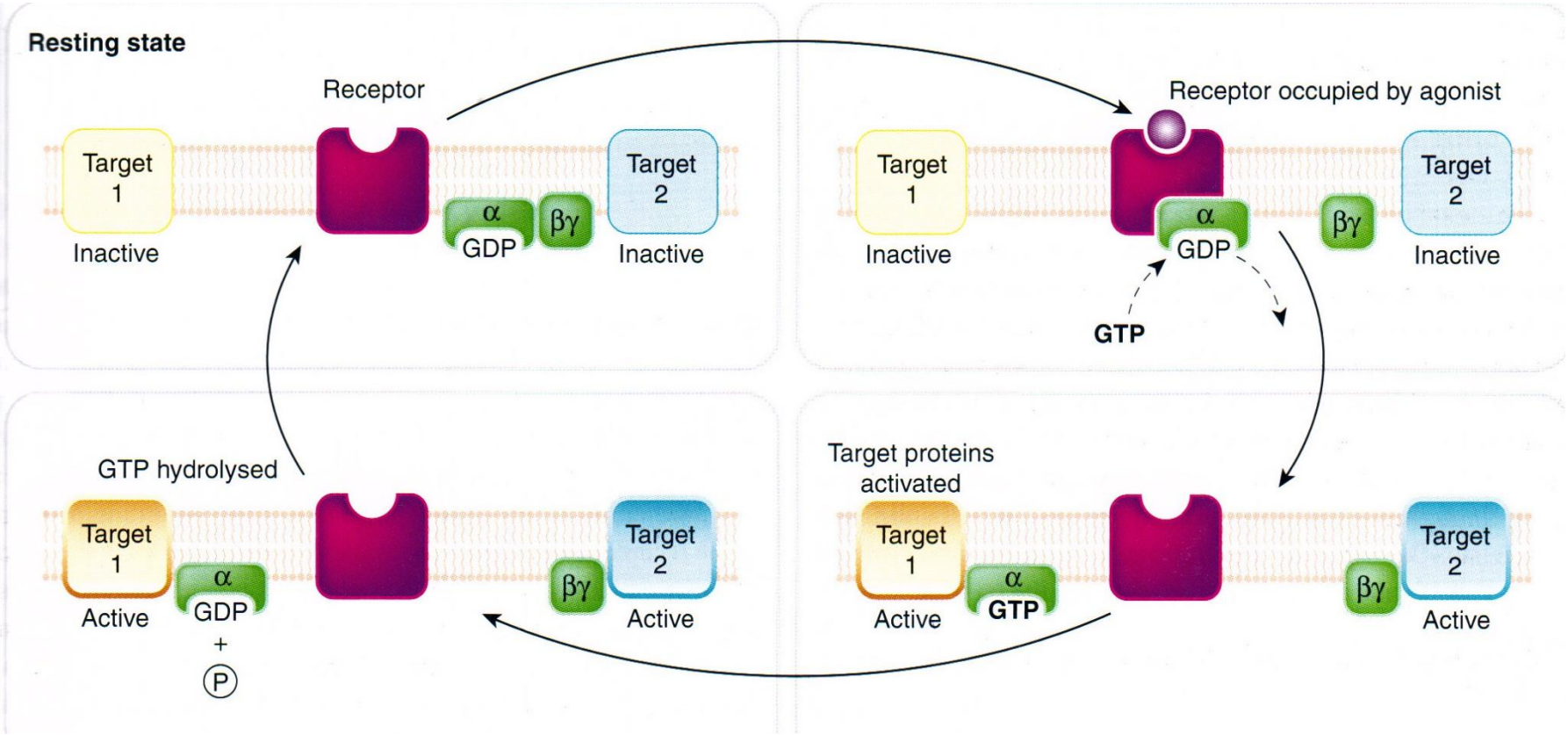


Fig. The function of the G-protein.
The G-protein consists of three subunits (α, β, γ), which are anchored to the membrane through attached lipid residues. Coupling of the α subunit to an agonist-occupied receptor causes the bound GDP to exchange with intracellular GTP; the α-GTP complex then dissociates from the receptor and from the βγ complex, and interacts with a target protein (target 1, which may be an enzyme, such as adenylate cyclase, or an ion channel). The βγ complex may also activate a target protein (target 2). The GTPase activity of the α subunit is increased when the target protein is bound, leading to hydrolysis of the bound GTP to GDP, whereupon the α subunit reunites with βγ.

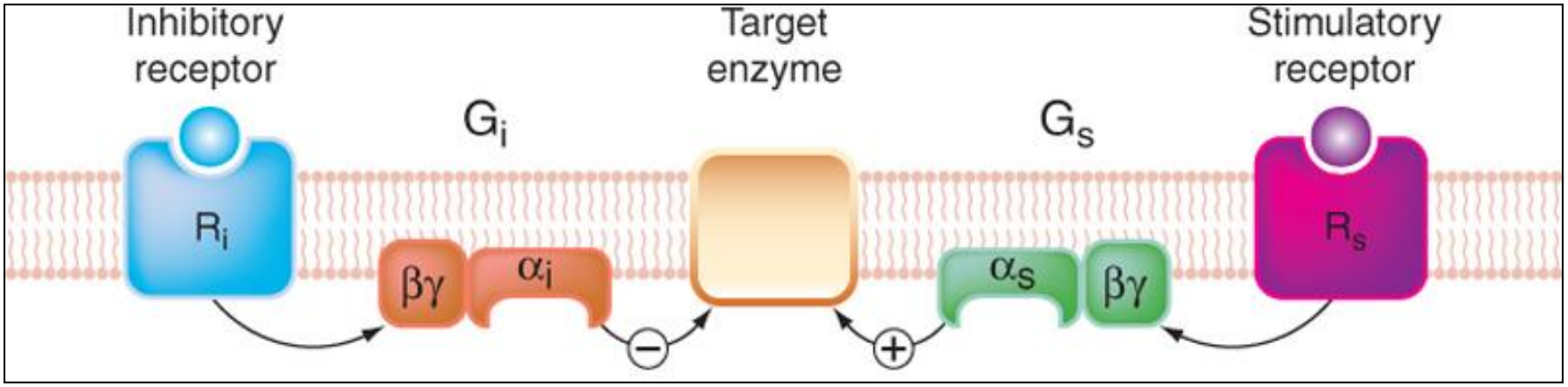


Figure. Bidirectional control of a target enzyme, such as adenylated cyclase by G_s and G_i. Heterogeneity of G-proteins allows different receptors to exert opposite effects on a target enzyme.

The main G-protein subtypes and their functions

Subtypes	Main effectors	Notes
Gα subunits		
$G\alpha_s$	Stimulates adenylyl cyclase, causing increased cAMP formation	Activated by cholera toxin, which blocks GTPase activity, thus preventing inactivation
$G\alpha_i$	Inhibits adenylyl cyclase, decreasing cAMP formation	Blocked by pertussis toxin, which prevents dissociation of αβγ complex
$G\alpha_o$	?Limited effects of α subunit (effects mainly due to βγ subunits)	Blocked by pertussis toxin. Occurs mainly in nervous system
$G\alpha_q$	Activates phospholipase C, increasing production of second messengers inositol triphosphate and diacylglycerol thus releasing Ca^{2++} from intracellular stores and activating protein kinase C (PKC)	
$G\alpha_{12/13}$	Activates Rho and thus Rho kinase	
Gβγ subunits	Activate potassium channels Inhibit voltage-gated calcium channels Activate GPCR kinases Activate mitogen-activated protein kinase cascade Interact with some forms of adenylyl cyclase and with phospholipase Cβ	Many βγ isoforms identified, but specific functions are not yet known

Effectors controlled by G-proteins

Two key second messenger pathways are controlled by receptors via G-proteins.

- **Adenylyl cyclase/cAMP:**
  - **can be activated or inhibited by pharmacological ligands, depending on the nature of the receptor and G protein;**
  - **adenylyl cyclase catalyses formation of the intracellular messenger cAMP;**
  - **cAMP activates protein kinases such as protein kinase A(PKA) that control cell function in many different ways by causing phosphorylation of various enzymes, carriers and other proteins.**

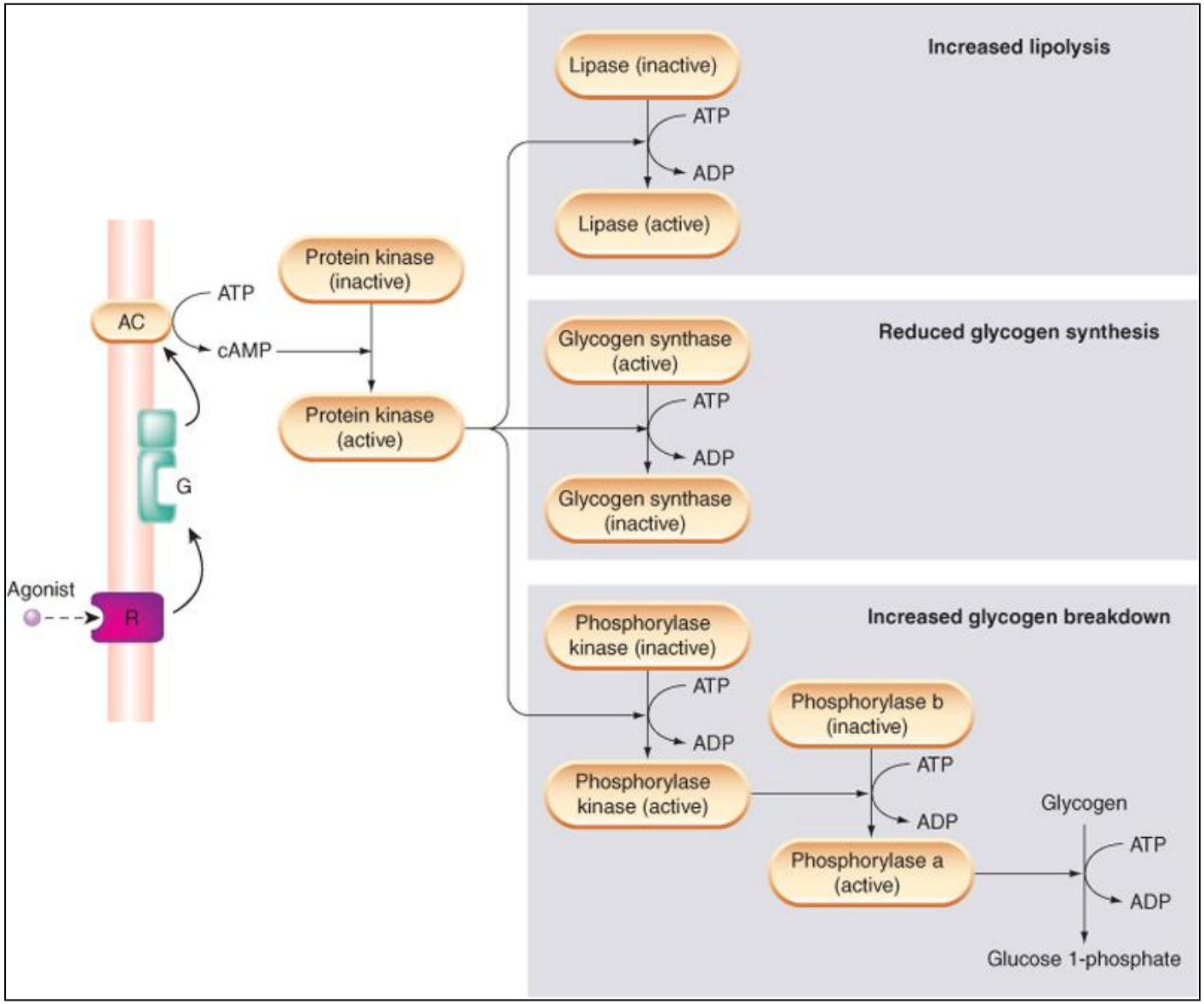


Fig. Regulation of energy metabolism by cAMP. AC, adenylate cyclase.

Effectors controlled by G-proteins

- **Phospholipase C/inositol trisphosphate (IP_3)/diacylglycerol (DAG):**
  - **catalyses the formation of two intracellular messengers, IP_3 and DAG, from membrane phospholipid;**
  - **IP_3 acts to increase free cytosolic Ca^{2+} by releasing Ca^{2+} from intracellular compartments**
  - **increased free Ca^{2+} initiates many events, including contraction, secretion, enzyme activation and membrane hyperpolarization;**
  - **DAG activates various protein kinase C (PKC) isoforms, which controls many cellular functions by phosphorylating a variety of proteins.**

Effectors controlled by G-proteins

Receptor-linked G-proteins also control:

- **Ion channels:**
  - **opening potassium channels, resulting in membrane hyperpolarization;**
  - **Inhibiting calcium channels, thus reducing neurotransmitter release.**
- **Phospholipase A_2 (and thus the formation of arachidonic acid and eicosanoids).**

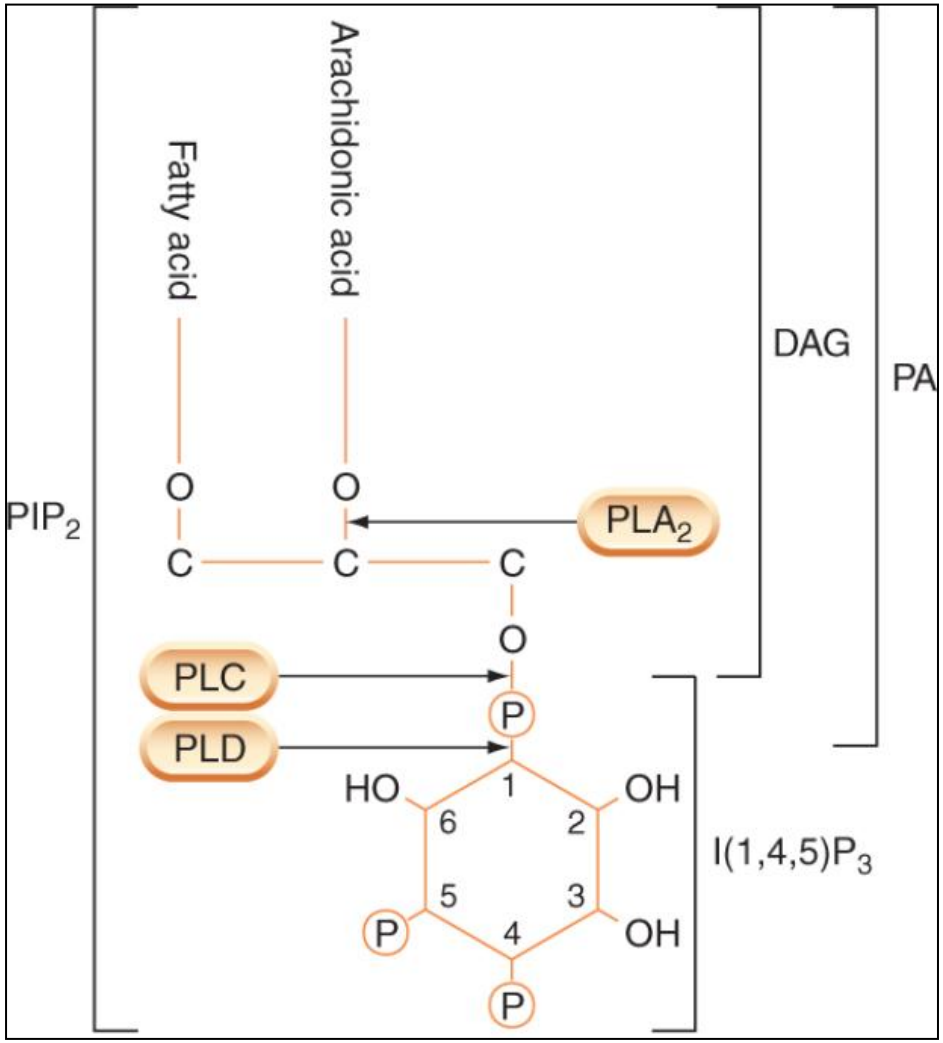


Fig. Structure of phosphatidylinositol bisphosphate (PIP2), showing sites of cleavage by different phospholipases to produce active mediators.

Cleavage by phospholipase A_2 (PLA_2) yields arachidonic acid. Cleavage by phospholipase C (PLC) yields inositol trisphosphate ($I(1,4,5)P_3$) and diacylglycerol (DAG). PA, phosphatidic acid; PLD, phospholipase D.

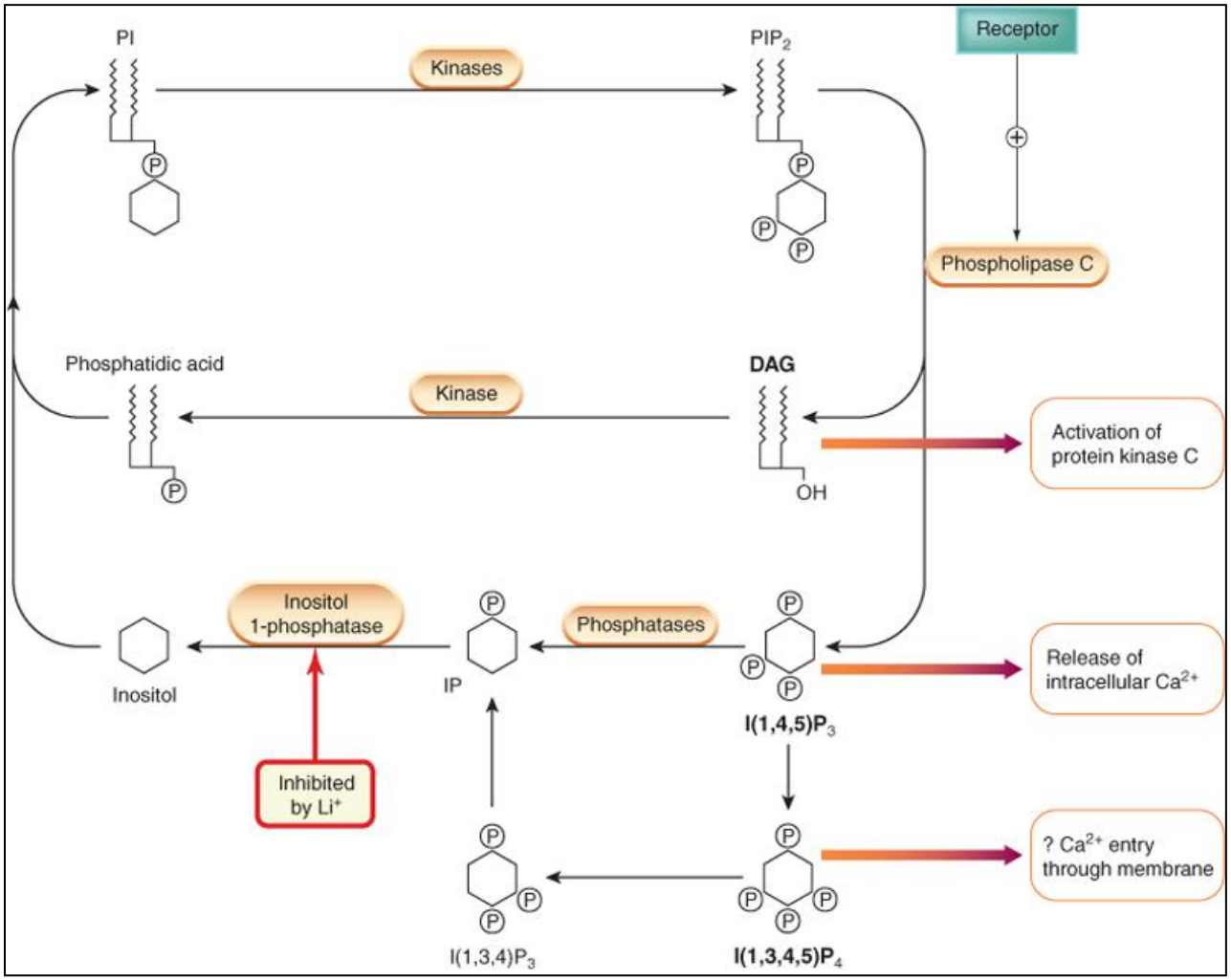


Fig. The phosphatidylinositol (PI) cycle.
Receptor-mediated activation of phospholipase C results in the cleavage of phosphatidylinositol biphosphate (PIP_2), forming diacylglycerol (DAG) (which activates protein kinase C) and inositol trisphosphate (IP_3) (which releases intracellular Ca^{2+}). The role of inositol tetraphosphate (IP_4), which is formed from IP_3 and other inositol phosphates, is unclear, but it may facilitate Ca^{2+} entry through the plasma membrane. IP_3 is inactivated by dephosphorylation to inositol. DAG is converted to phosphatidic acid, and these two products are used to regenerate PI and PIP_2.

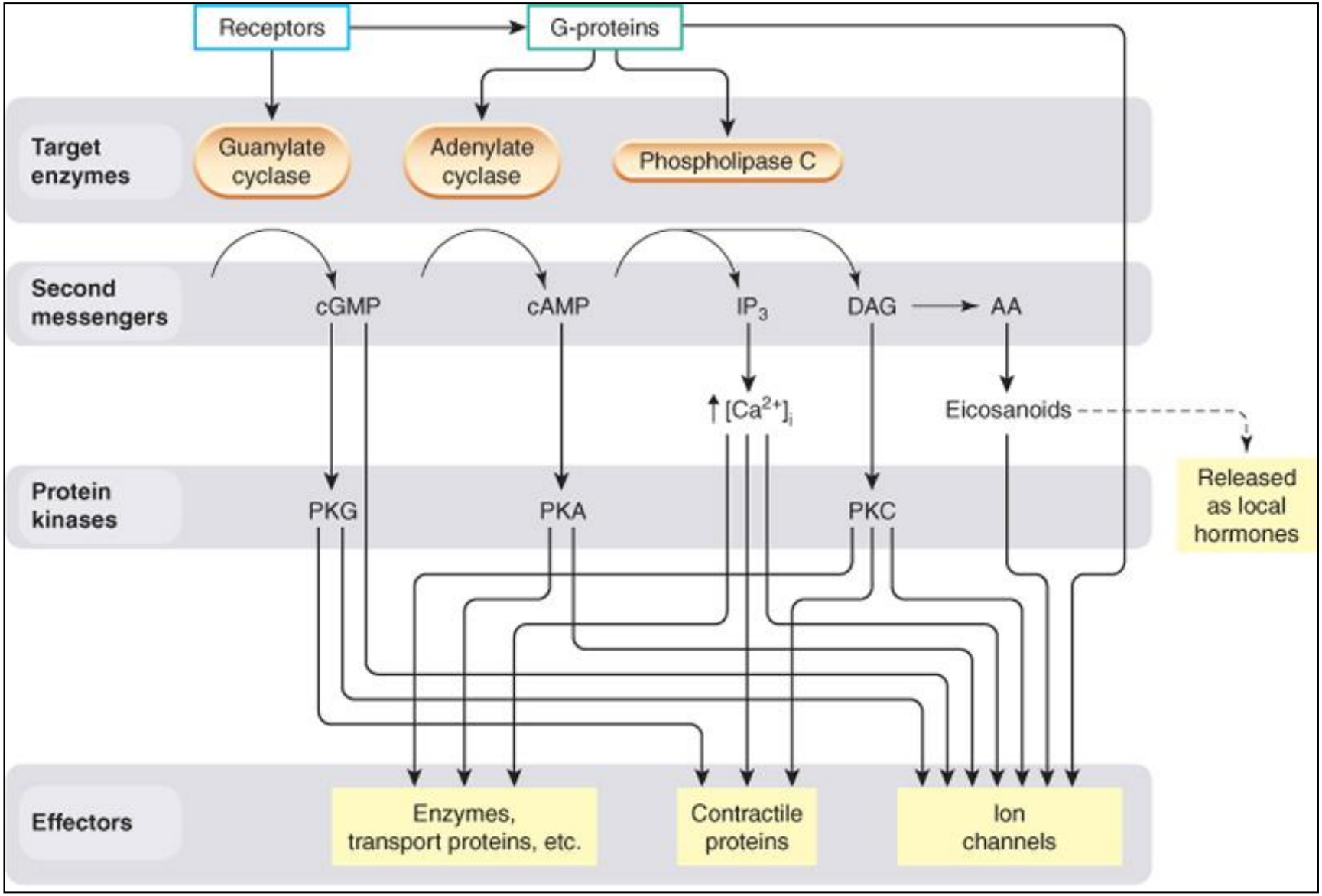


Fig. G-protein and second messenger control of cellular effector systems.
AA, arachidonic acid; DAG, diacylglycerol; IP_3, inositol trisphosphate.

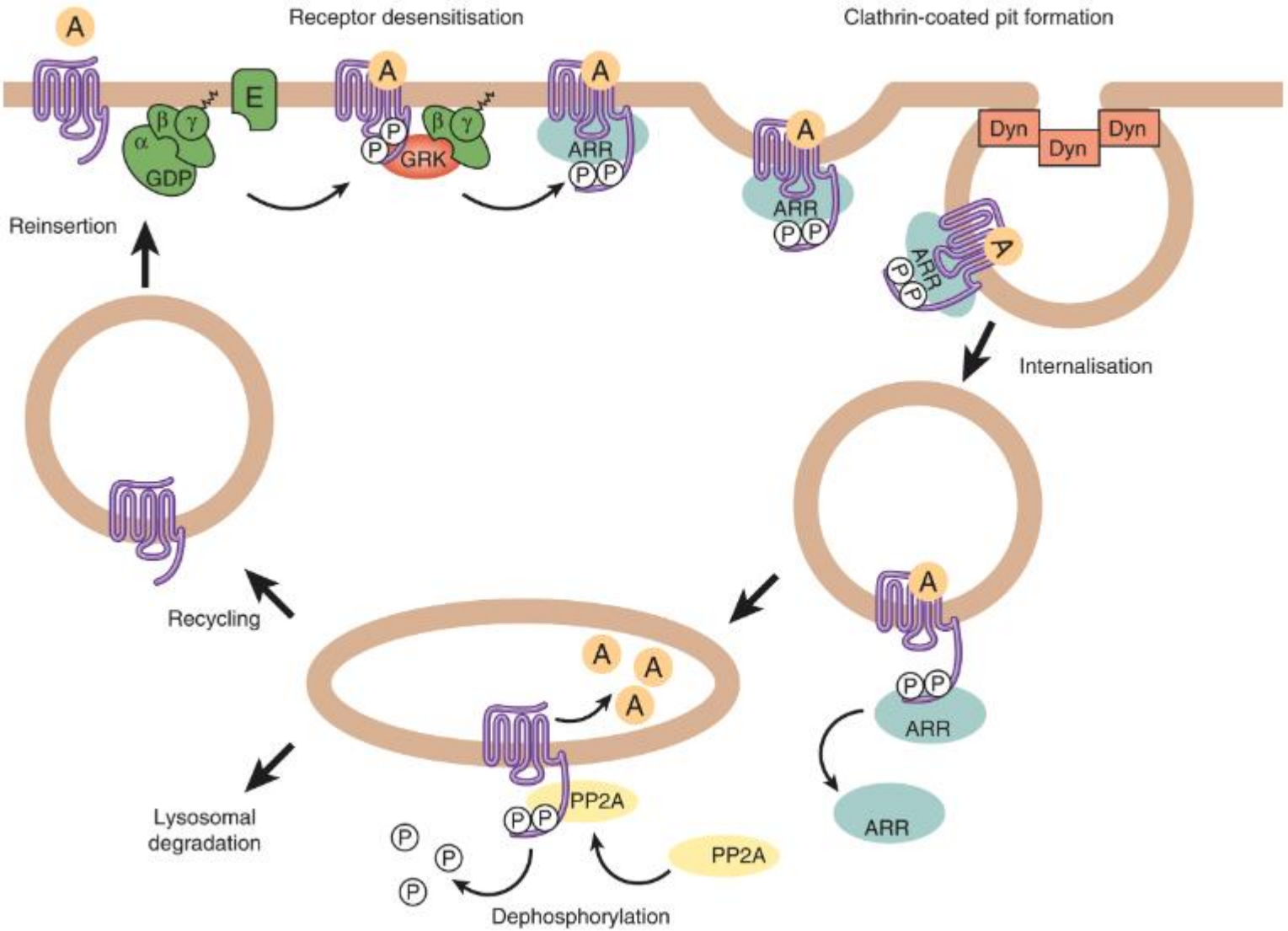


Fig. Desensitisation and trafficking of G protein-coupled receptors (GPCRs).

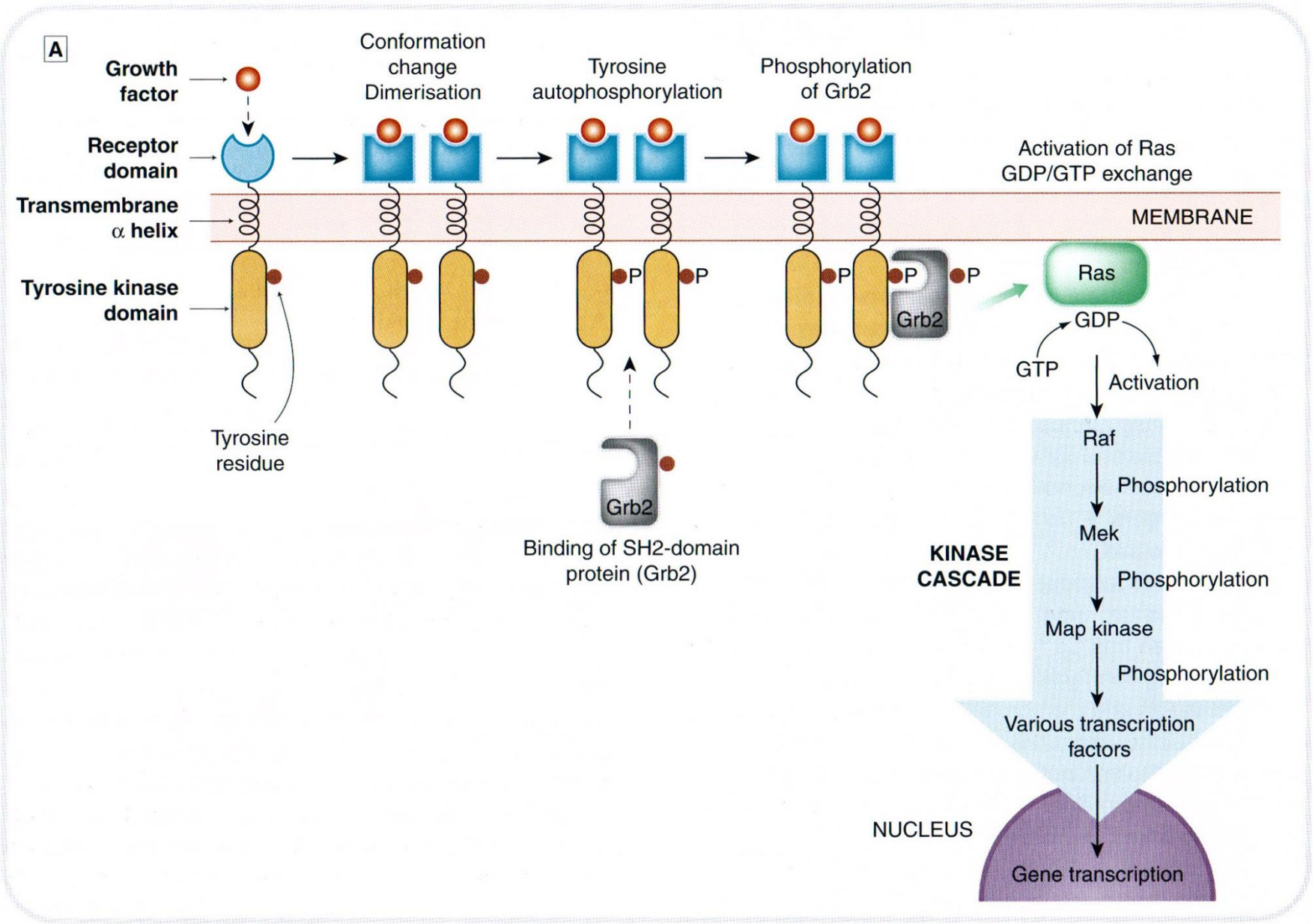
On prolonged agonist activation of the GPCR selective G protein-coupled receptor kinases (GRKs) are recruited to the plasma membrane and phosphorylate the receptor. Arrestin (ARR) then binds and trafficks the GPCR to clathrin-coated pits for subsequent internalisation into endosomes in a dynamin-dependent process. The GPCR is then dephosphorylated by a phosphatase (PP2A) and either recycled back to the plasma membrane or trafficked to lysosomes for degradation. ARR, arrestin; Dyn, dynamin; GRK, G protein-coupled receptor kinase; PP2A, phosphatase 2A.

Kinase-linked receptors

- **Receptors for various growth factors incorporate tyrosine kinase in their intracellular domain.**
- **Cytokine receptors have an intracellular domain that binds and activates cytosolic kinases when the receptor is occupied.**
- **The receptors all share a common architecture, with a large extracellular ligand-binding domain connected via a single membrane-spanning helix to the intracellular domain.**
- **Signal transduction generally involves dimerisation of receptors, followed by autophosphorylation of tyrosine residues. The phosphotyrosine residues act as acceptors for the SH2 domains of a variety of intracellular proteins, thereby allowing control of many cell functions.**

Kinase-linked receptors

- **They are involved mainly in events controlling cell growth and differentiation, and act indirectly by regulating gene transcription.**
- **Two important pathways are:**
  - **the Ras/Raf/mitogen-activated protein (MAP) kinase pathway, which is important in cell division, growth and differentiation**
  - **the Jak/Stat pathway activated by many cytokines, which controls the synthesis and release of many inflammatory mediators.**


A
Growth factor
Receptor domain
Transmembrane α helix
Tyrosine kinase domain
Tyrosine residue
Conformation change Dimerisation
Tyrosine autophosphorylation
Phosphorylation of Grb2
Grb2
Binding of SH2-domain protein (Grb2)
Activation of Ras GDP/GTP exchange
MEMBRANE
P
Grb2
Ras
GDP
GTP
Activation
Raf
Phosphorylation
Mek
Phosphorylation
Map kinase
Phosphorylation
KINASE CASCADE
Various transcription factors
NUCLEUS
Gene transcription

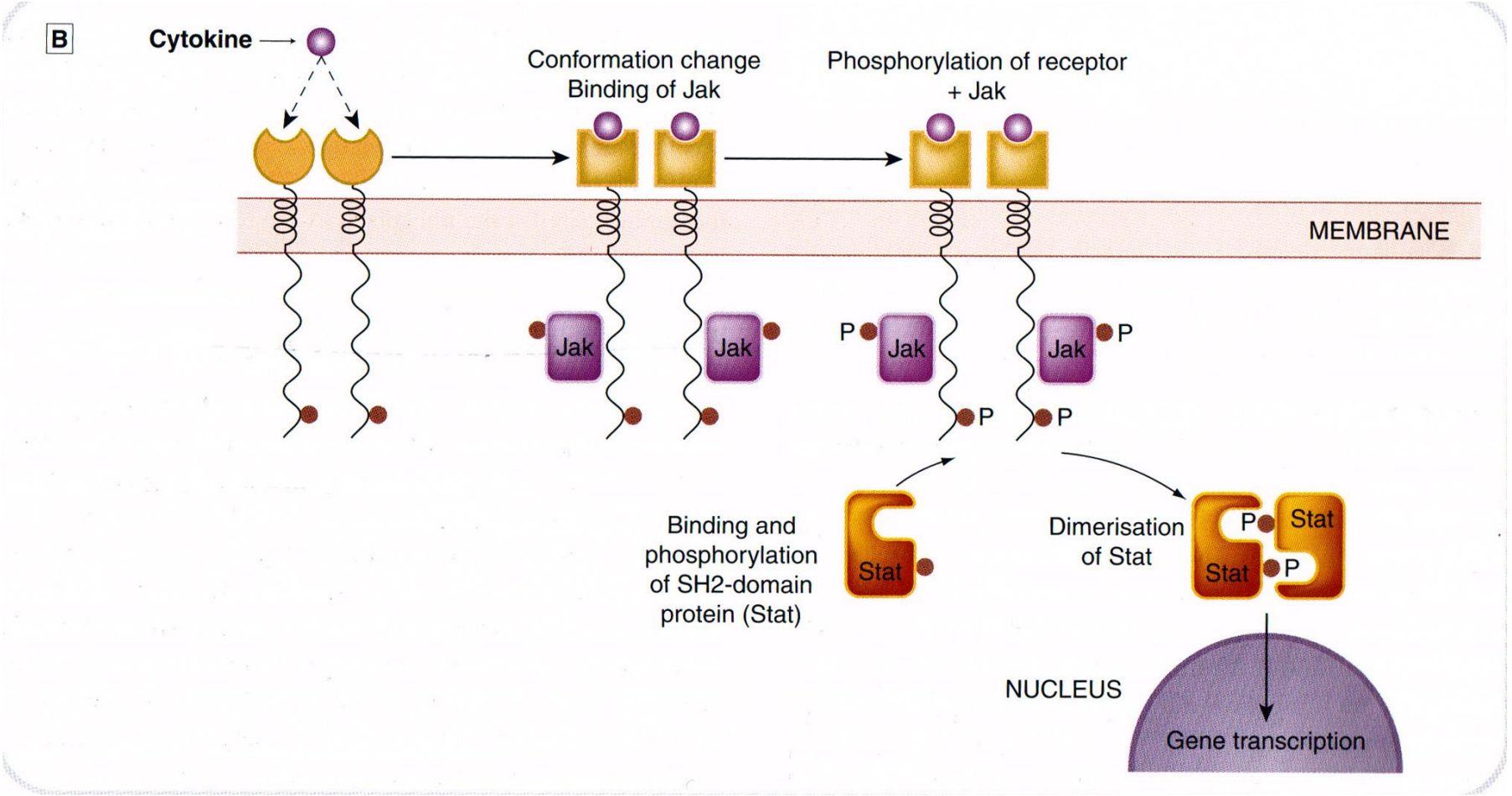


Fig. Transduction mechanisms of kinase-linked receptors.
The first step following agonist binding is dimerisation, which leads to autophosphorylation of the intracellular domain of each receptor. SH2 domain proteins then bind to the phosphorylated receptor and are themselves phosphorylated. Two well-characterised pathways are shown: [A] The growth factor (Ras/Raf/mitogen-activated protein [MAP] kinase) pathway; [B] the cytokine (Jak/Stat) pathway. Several other pathways exist, and these phosphorylation cascades interact with components of G-protein systems.

Protein phosphorylation in signal transduction

- **Many receptor-mediated events involve protein phosphorylation, which controls the functional and binding properties of intracellular proteins.**
- **Receptor-linked tyrosine kinases, cyclic nucleotide-activated tyrosine kinases, and intracellular serine/threonine kinases comprise a 'kinase cascade' mechanism that leads to amplification of receptor-mediated events.**
- **There are many kinases, with differing substrate specificities, allowing specificity in the pathways activated by different hormones.**
- **Desensitization of G-protein-coupled receptors occurs as a result of phosphorylation by specific receptor kinases, causing the receptor to become non-functional and to be internalized.**
- **There is a large family of phosphatases that act to dephosphorylate proteins and thus reverse the effects of kinases.**

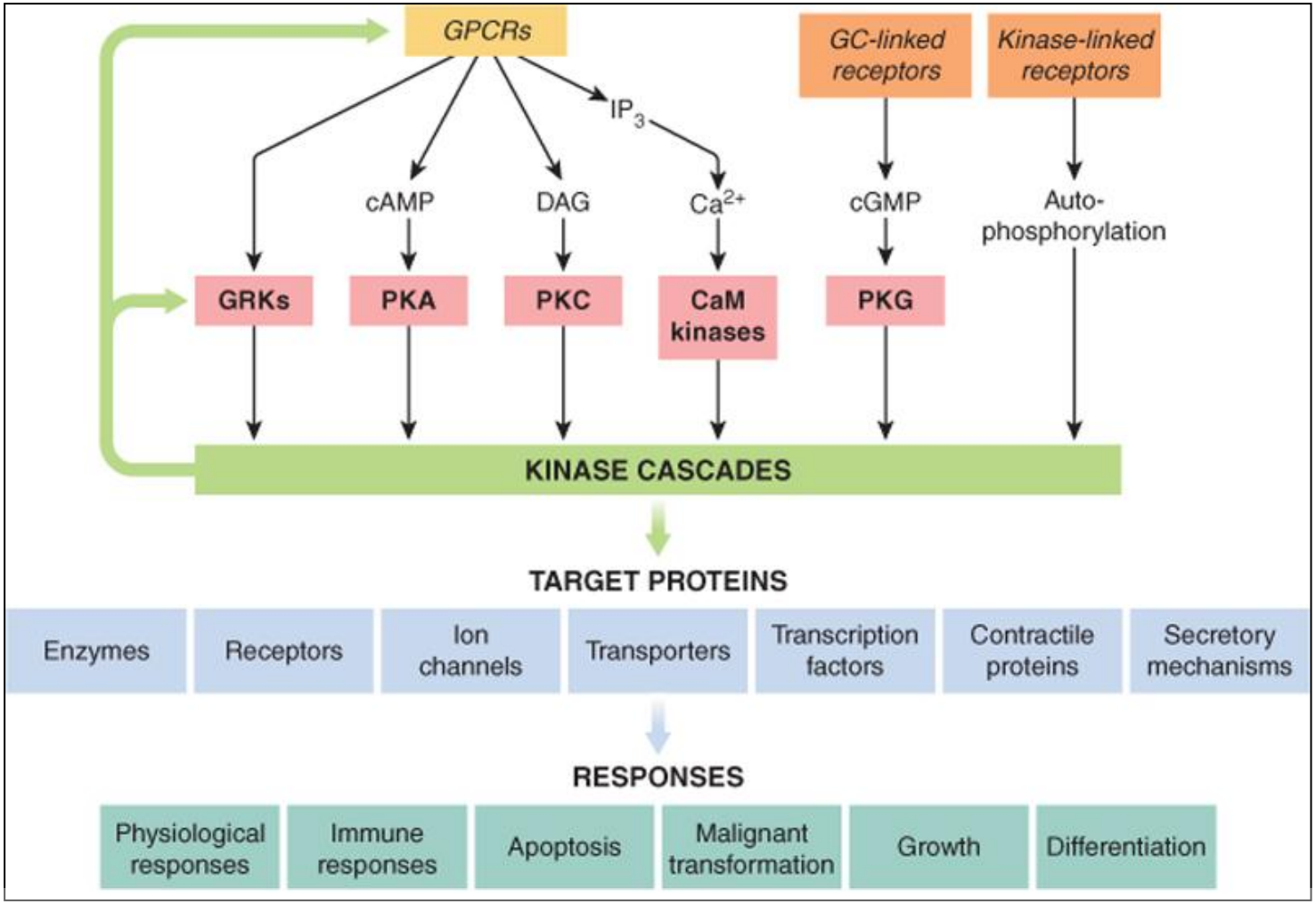


Fig. Central role of kinase cascades in signal transduction.

Kinase cascades are activated by GPCRs, either directly or via different second messengers, by receptors that generate cGMP, or by kinase-linked receptors. The kinase cascades regulate various target proteins, which in turn produce a wide variety of short- and long-term effects. CaM kinase, Ca^{2+}/calmodulin-dependent kinase; DAG, diacylglycerol; GC, guanylate cyclase; GRK, GPCR kinase; IP^3, inositol trisphosphate; PKA, cAMP-dependent protein kinase; PKC, protein kinase C; PKG, cGMP-dependent protein kinase.

Nuclear receptors

- A family of 48 soluble receptors that sense lipid and hormonal signals and modulate gene transcription.

- Their ligands are many and varied, including steroid drugs and hormone, thyroid hormones, vitamins A and D, various lipids and xenobiotics.

- There are two main categories:

  - Class I NRs are present in the cytoplasm, form homodimers in the presence of their ligand, and migrate to the nucleus. Their ligands are mainly endocrine in nature (e.g. steroid hormones).
  - Class II NRs are generally constitutively present in the nucleus and form heterodimers with the retinoid X receptor. Their ligands are usually lipids (e.g. the fatty acids).

Nuclear receptors

- **The liganded receptor complexes initiate changes in gene transcription by binding to hormone response elements in gene promoters and recruiting co-activator or co-repressor factors.**

- **The receptor family is the target of approximately 10% of prescription drugs, and the enzymes that it regulates affect the pharmacokinetics of some 60% of all prescription drugs.**

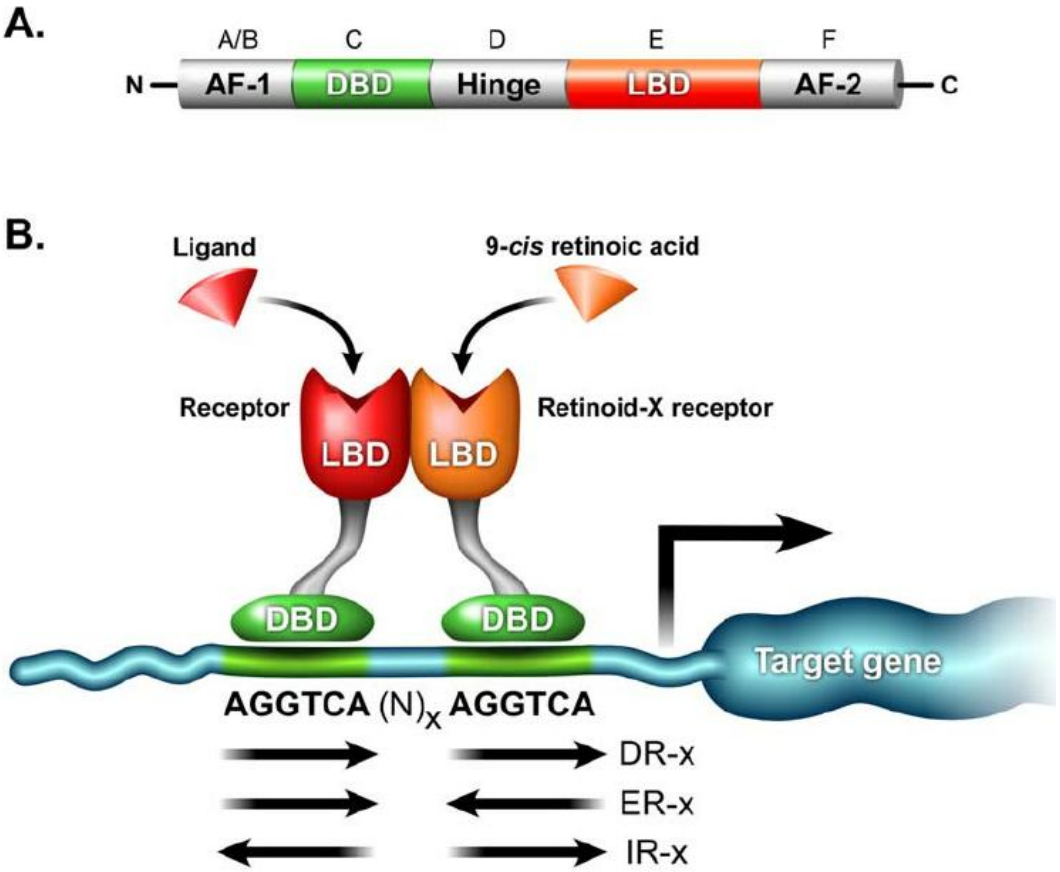


Fig. Structure and DNA binding of nuclear receptors. (A) Schematic diagram for a common domain structure of NRs which include N-terminal activation function 1(AF-1), DNA binding domain (DBD) consisting of two zinc fingers (ZF), hinge region (Hinge), ligand binding domain (LBD), and C-terminal AF-2. (B) Schematic diagram for NR dimerization and DNA binding. Nuclear receptors bind as heterodimers with RXR to repeats of the nucleotide hexamer AGGTCA with variable spacing. The hexamers can be arranged either as direct repeats (DR), everted repeats (ER), or inverted repeats(IR).